

GNN-Based BERT for Understanding Context from Music

Course: Neural Networks CSE425

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Abstract

This project builds a hybrid GNN-BERT system for understanding musical context, combining chord-transition graph structure with lyric and caption semantics across four tasks: (1) a BERT baseline predicting genre from lyrics, (2) a GraphSAGE GNN on chord-transition graphs compared against a CNN mel-spectrogram baseline, (3) a GNN BERT fusion model with four ablation variants, and (4) a contrastive dual-encoder for cross-modal audio-caption retrieval using MusicCaps. Experiments were run on the GTZAN dataset (8 genres, classical and jazz excluded due to chord-template incompatibility) and a MusicCaps subset. The CNN mel-spectrogram baseline substantially outperformed the chord-graph GNN (72.4% vs 32.7% test accuracy), while the GNN-BERT cross-attention fusion model outperformed both individual modalities in the Task 3 ablation study (42.2% test accuracy). The contrastive retrieval model in Task 4 achieved modest but above-chance retrieval performance after a hyperparameter correction. Results, limitations, and error analysis are discussed for each task.

1. Introduction and Motivation

Music is a multi-layered signal spanning melody, harmony, rhythm, lyrics, and listener-described semantics. This project investigates whether combining a Graph Neural Network (GNN) encoding harmonic/structural information with a BERT-based text encoder can better capture musical context than either modality alone. Four tasks were implemented, of increasing complexity: a text-only BERT baseline, a graph-only GNN model (with a CNN baseline for comparison), a GNN-BERT fusion model with ablations, and a cross-modal contrastive retrieval system.

2. Dataset and Preprocessing

2.1 GTZAN Dataset

The GTZAN dataset was used for Tasks 1-3. Of the original 10 genres, classical and jazz were excluded because their harmonic complexity (frequent key changes, extended/altered chords, ambiguous tonality) is poorly suited to the simple major/minor triad template-matching approach used for chord recognition in this project. The remaining 8 genres used were: blues, country, disco, hip-hop, metal, pop, reggae, and rock.

Lyrics were extracted via OpenAI Whisper (base model, CPU inference) applied to each audio clip. Since GTZAN clips are 30-second excerpts and several genres are substantially or fully instrumental,

transcription quality and coverage varied by genre (see Section 2.4).

2.2 Chord-Transition Graph Construction

Each 30-second audio clip was converted into a chord-transition graph through a four-stage pipeline:

- Stage A: Chroma feature extraction using librosa's `chroma_cqt` at 0.5-second window resolution (60 frames per 30s clip).
- Stage B: Chord labeling via cosine-similarity template matching against 24 chord templates (12 major and 12 minor triads over the 12 pitch classes).
- Stage C: Smoothing via median filtering (window = 9 frames, 4.5s) to remove frame-level flickering, followed by run-length encoding into chord segments, with a minimum segment duration of 1.0s (segments shorter than this were merged into a neighboring segment).
- Stage D: Graph construction — nodes represent unique chords present in the track (features = average chroma vector across all occurrences of that chord), and edges represent observed chord-to-chord transitions, weighted by transition frequency.

These smoothing parameters were empirically tuned: an initial pass (window=15, min duration=1.5s) over-smoothed tracks down to as few as 1-2 nodes; the final parameters (window=9, min duration=1.0s) were validated on two independent 24-track samples (3 tracks per genre), achieving 88-96% of sampled tracks within the target range of 3- 8 unique chord nodes.

2.3 Degenerate Graph Filtering

Tracks producing graphs with fewer than 3 nodes or fewer than 3 edges were excluded from GNN training, as such graphs provide insufficient structure for GraphSAGE message passing. Across the full extraction run, 35 of 400 tracks (8.75%) were excluded, fairly evenly distributed across genres (reggae and blues showing the highest exclusion counts, consistent with their tendency toward sparse or droning chord patterns). The final GNN/CNN baseline dataset (Task 2) comprised 365 usable tracks.

2.4 Lyrics Coverage and Fusion-Ready Dataset

For Task 3, tracks were required to have both a valid chord graph and non-trivial lyrics (5+ transcribed words) to be usable. Coverage analysis across an expanded 800-track pool (100 per genre) found 678 tracks with usable lyrics and 751 tracks with valid graphs; the intersection yielded 635 fusion-ready tracks (79.4% of the pool), reasonably balanced across genres (69-91 tracks per genre).

2.5 MusicCaps Dataset (Task 4)

For the cross-modal contrastive task, 300 MusicCaps (audio, caption) pairs were sampled and their corresponding 10-second audio clips downloaded from YouTube via `yt-dlp`; 278 clips were successfully retrieved (92.67% success rate). The same chord-graph extraction pipeline was reapplied with adjusted parameters suited to the shorter clip length (median filter window = 5 frames, minimum segment duration = 0.5s). After excluding 7 degenerate graphs (2.52%), 271 usable (graph, caption) pairs remained for contrastive training.

3. Task 1: BERT Baseline (Lyrics to Genre)

3.1 Method

A bert-base-uncased model was fine-tuned with a linear classification head on the CLS token embedding, predicting genre from transcribed lyrics (max sequence length 256 tokens). Training used early stopping on validation loss (patience = 3 epochs).

3.2 Results

Metric	Value
Test Accuracy	23.96%
Macro F1	0.1970
Micro F1	0.2396
Best epoch	5 (of 5, early stopping not triggered)

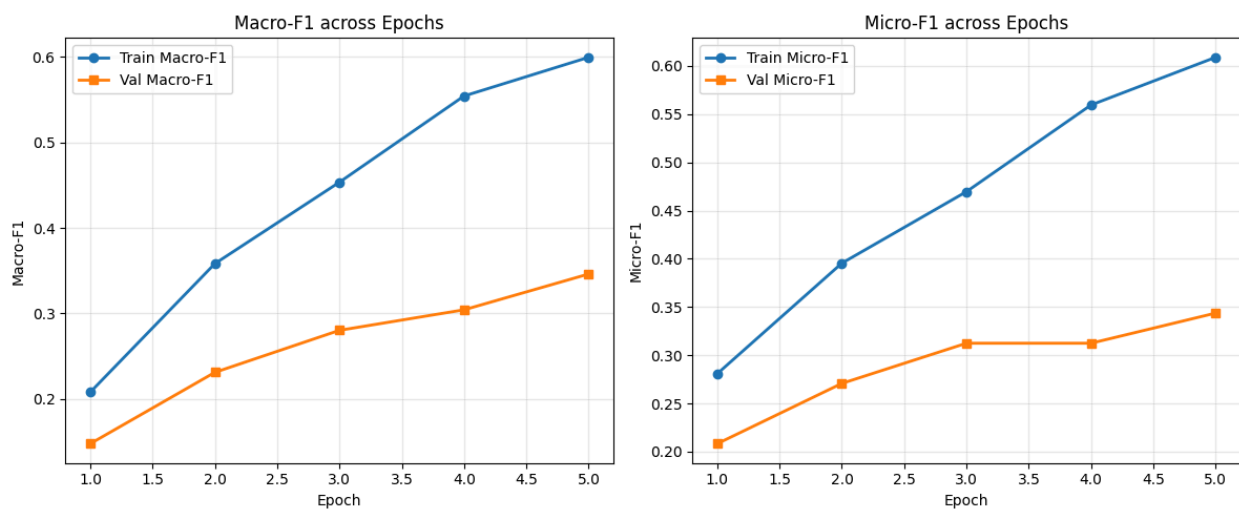


FIGURE: Task 1 F1 curves across epochs

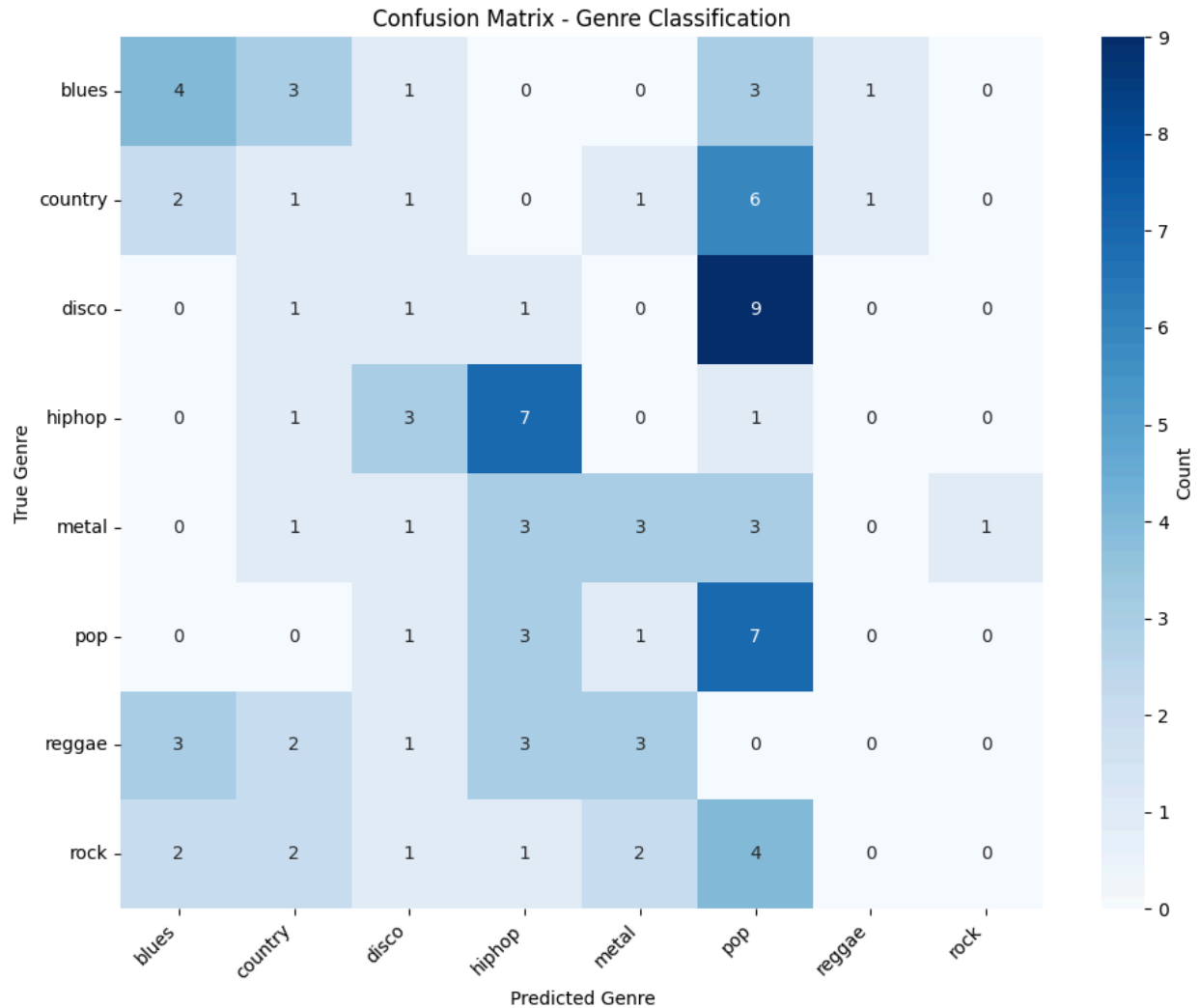


FIGURE: Task 1 confusion matrix

3.3 Discussion

The model shows clear overfitting: training accuracy reached 60.9-67.6% while validation accuracy plateaued near 34-35% and test accuracy landed lower still at 23.96%. This gap is attributable to two compounding factors: (1) the small dataset size (under 500 training examples) relative to BERT's parameter count, and (2) noise in the lyrics themselves, since they are ASR-derived (Whisper) transcriptions rather than ground-truth lyrics, and several genres are substantially instrumental, yielding sparse or hallucinated transcriptions. Qualitative inspection of test predictions also revealed a bias toward predicting "metal" across multiple genuinely different genres, suggesting the model may have latched onto superficial lexical cues (e.g. aggressive word choice) rather than genre-specific semantic patterns. Despite these limitations, the result (23.96%) remains meaningfully above the 12.5% random chance baseline for 8-class classification.

4. Task 2: GNN on Chord-Transition Graphs, with CNN Baseline

4.1 Method

A 2-layer GraphSAGE encoder processed each track's chord-transition graph (12-dimensional chroma node features), followed by global mean pooling and a linear classification head, trained with cross-entropy loss. As a baseline, a 4-block CNN (Conv2D-BatchNorm-ReLU-MaxPool, channels 1-16-32-64-128) was trained on full log mel spectrograms (128 mel bins) for the same 365-track dataset and identical train/val/test split.

4.2 Results

Model	Test Accuracy	Notes
GraphSAGE (chord graph)	32.69%	Best val epoch checkpoint
CNN (mel-spectrogram)	72.37%	Best val epoch checkpoint

Per-genre F1 scores (GraphSAGE): blues 0.29, country 0.38, disco 0.43, hiphop 0.25, metal 0.44, pop 0.31, reggae 0.31, rock 0.00 (rock was never correctly predicted).

Per-genre F1 scores (CNN): blues 0.63, country 0.57, disco 0.82, hiphop 0.89, metal 0.95, pop 0.82, reggae 0.58, rock 0.53.

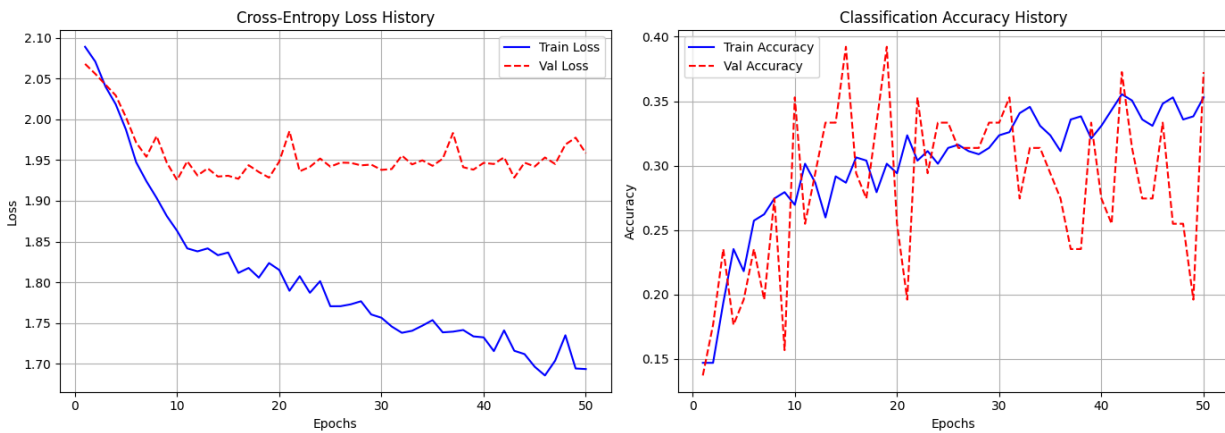


FIGURE: GNN training curve

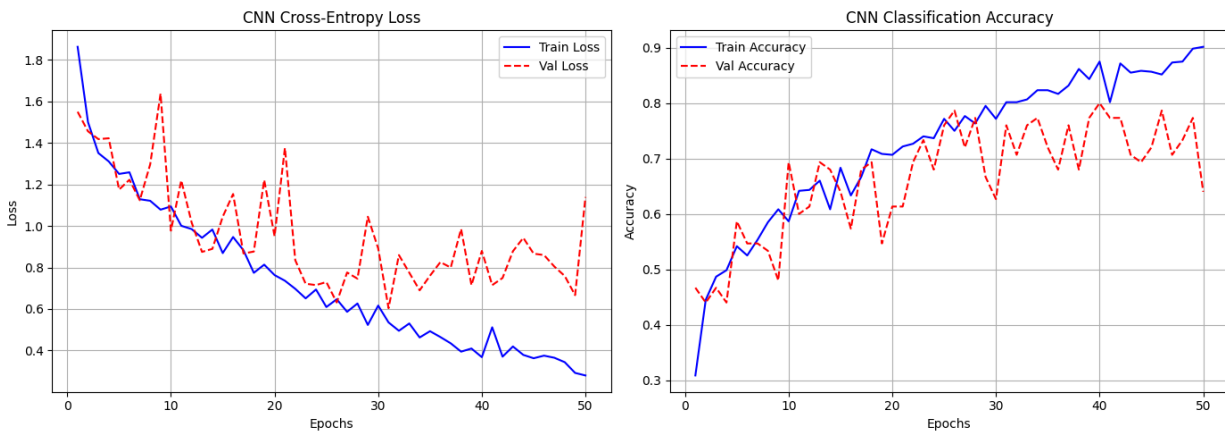


FIGURE: CNN training curve

4.3 Discussion

The CNN baseline substantially outperformed the chord-graph GNN. This is an expected and informative finding: mel-spectrograms retain timbre, instrumentation, rhythm, and production-style information, all strong genre discriminators, whereas the chord-transition graph captures only harmonic movement, distilled down to as few as 3- 8 nodes per 30-second clip. Harmony alone is a comparatively weak and noisy genre signal, particularly for genres with overlapping chord vocabularies (e.g. rock, blues, and country frequently share simple I-IV-V progressions), which likely explains the GNN's complete failure on rock ($F1 = 0.00$). This comparison directly supports the project's motivating premise that richer, multi-modal representations are needed to fully capture musical context, and sets up the Task 3 fusion experiment.

5. Task 3: GNN-BERT Fusion

5.1 Method

Four model variants were trained and compared on the same 635-track fusion-ready dataset (80/10/10 stratified split): (1) GNN-only (ChordGraphSAGE + linear head), (2) BERT-only (BertLyricsEncoder CLS token + linear head), (3) Concat-fusion (graph embedding g concatenated with text embedding t), and (4) Cross-attention fusion (graph embedding g as query attending over BERT's full token-level hidden states as keys/values). BERT parameters were partially frozen (all but the last 2 transformer layers) in all variants to reduce overfitting risk.

5.2 Ablation Results

Model Variant	Test Accuracy	Macro F1	Micro F1
GNN-only	24.44%	0.1552	0.2444
BERT-only	40.00%	0.3844	0.4000
Concat-fusion	37.78%	0.3507	0.3778
Cross-attention-fusion	42.22%	0.4225	0.4222

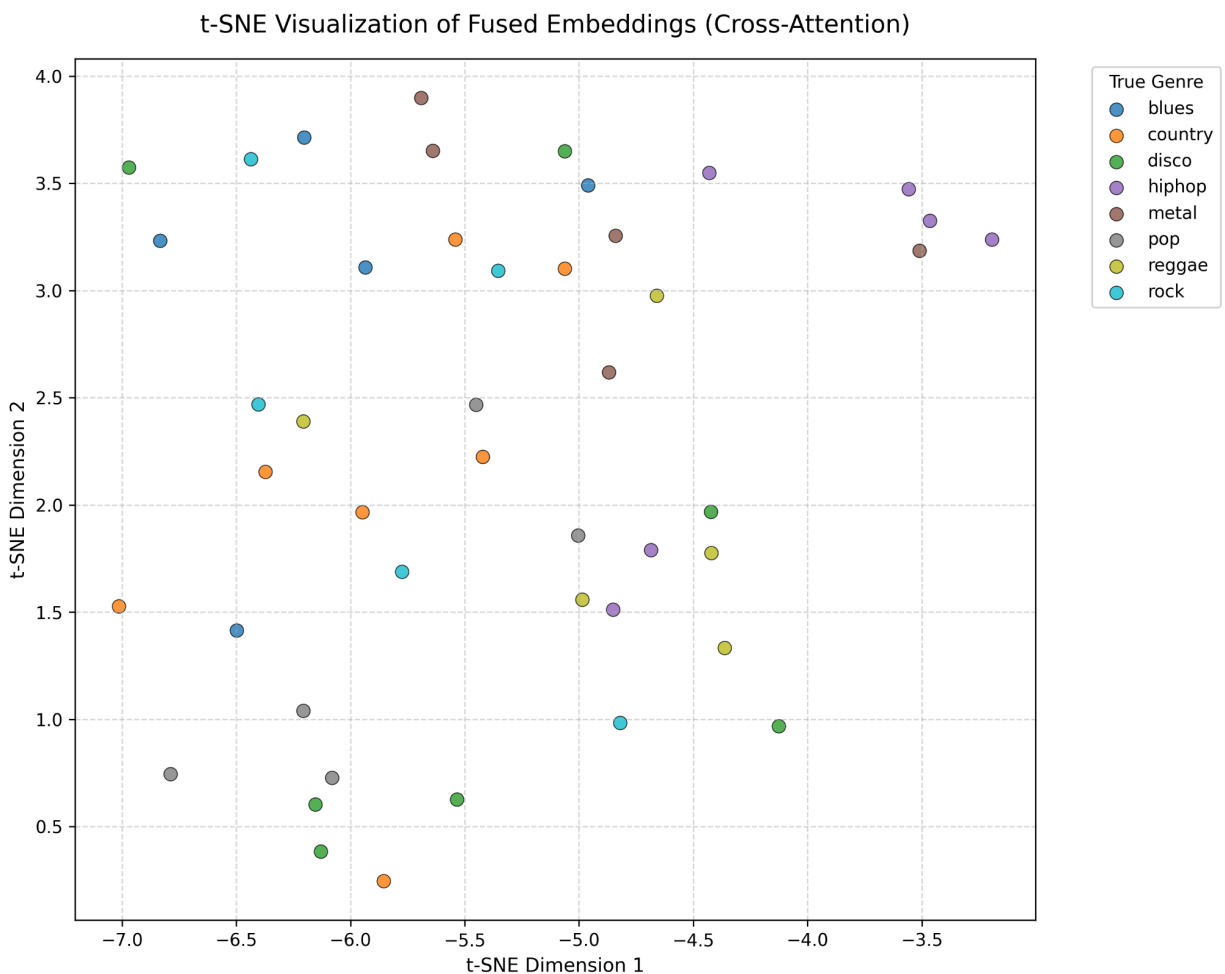


FIGURE: t-SNE of fused embeddings, colored by genre

5.3 Discussion

Cross-attention fusion achieved the best result across all four variants, outperforming both single-modality baselines and the simpler concatenation-based fusion, supporting the hypothesis that learned, weighted combination of graph and text representations captures genre-relevant information more effectively than either modality alone or naive concatenation.

However, the cross-attention model showed pronounced overfitting: by the final epoch, training accuracy reached 96.61% while validation accuracy was only 50.00%, and validation loss increased even as training loss fell sharply (0.16 vs 1.93). This is attributable to the small dataset size (635 tracks) relative to the combined GNN+BERT parameter count, even with partial BERT freezing. Test accuracy (42.22%) is nonetheless the best of the four variants and is reported as the primary result, with this overfitting noted as a limitation.

A striking consistent failure was observed for the reggae genre, which scored 0.00 precision, recall, and F1 across all four model variants. This suggests reggae tracks in this dataset are being systematically misclassified, likely absorbed into acoustically or lyrically similar genres such as disco or hip-hop, rather than reflecting a modeling artifact specific to any one architecture.

Note on dataset choice: the assignment specification recommends FMA-medium or MagnaTagATune for Task 3; GTZAN was used instead for consistency with Tasks 1-2 and due to its more tractable scale for CPU-only training.

5.4 Case Studies

Three representative test-set tracks were selected from the Cross-attention-fusion model's predictions to illustrate model behavior. Each case study pairs the track's chord-transition path with its transcribed lyrics and the model's prediction. Note: Task 3 does not include timestamp-level alignment between individual chords and specific lyric lines; graphs and lyrics are paired only at the track level.

Case Study 1: country.00021.wav — Correct Prediction (country)

Chord path: $A\# \rightarrow Am \rightarrow D \rightarrow C \rightarrow Fm$

Lyrics: 50 words transcribed (available in lyrics_transcriptions.csv).

The model correctly classified this track as a country. The chord path includes a major-to-relative-minor movement ($A\#$ to Am is not diatonically standard, but the overall progression's mix of major and minor triads with movement through D and C is consistent with common country harmonic patterns. The correct classification suggests the fusion model successfully leveraged either the harmonic pattern, lyrical content, or their combination for this track.

Case Study 2: metal.00021.wav — Correct Prediction

(metal) *Chord path:* $A \rightarrow Em \rightarrow E \rightarrow Gm$

Lyrics: 44 words transcribed (available in lyrics_transcriptions.csv).

The model correctly classified this track as metal. The presence of minor chords (Em , Gm) alongside power-chord like major/minor alternation is consistent with metal's typical harmonic minor-leaning character, suggesting the GNN branch likely contributed a useful signal here.

Case Study 3: pop.00018.wav — Incorrect Prediction (predicted country, true genre pop)

Chord path: $C\# \rightarrow Cm \rightarrow C\#m \rightarrow G\#$

Lyrics: 63 words transcribed (available in lyrics_transcriptions.csv).

The model misclassified this pop track as country. Given country's strong performance elsewhere in the dataset and pop's comparatively moderate F1 score in the ablation study, this may reflect lyrical or harmonic overlap between pop and country tracks in this particular dataset (both genres commonly feature simple major/minor progressions and narrative lyrical structures), illustrating a genuine ambiguity rather than a clear model failure.

6. Task 4: Cross-Modal Contrastive Retrieval (Bonus)

6.1 Method

A dual-encoder model was trained using the same ChordGraphSAGE and BertLyricsEncoder components as Tasks 2-3, each followed by a projection head to a shared 128-dimensional embedding space and L2 normalization. The model was trained with a symmetric InfoNCE contrastive loss on 271 (chord graph, caption) pairs from MusicCaps (80/10/10 split), using partial BERT fine-tuning.

6.2 Results

An initial training run converged prematurely, with the best validation loss occurring at epoch 1 and degrading thereafter — a sign of early overfitting or instability given the small dataset and batch size. Hyperparameters were adjusted (BERT learning rate reduced from $2e-5$ to $5e-6$, GNN/projection learning rate reduced from $1e-3$ to $5e-4$, temperature increased from 0.07 to 0.15, batch size reduced from 16 to 8, gradient clipping added), which improved results substantially:

Direction	R@1	R@5	R@10	MRR
Caption → Audio	3.57%	21.43%	39.29%	0.1675
Audio → Caption	3.57%	21.43%	39.29%	0.1675

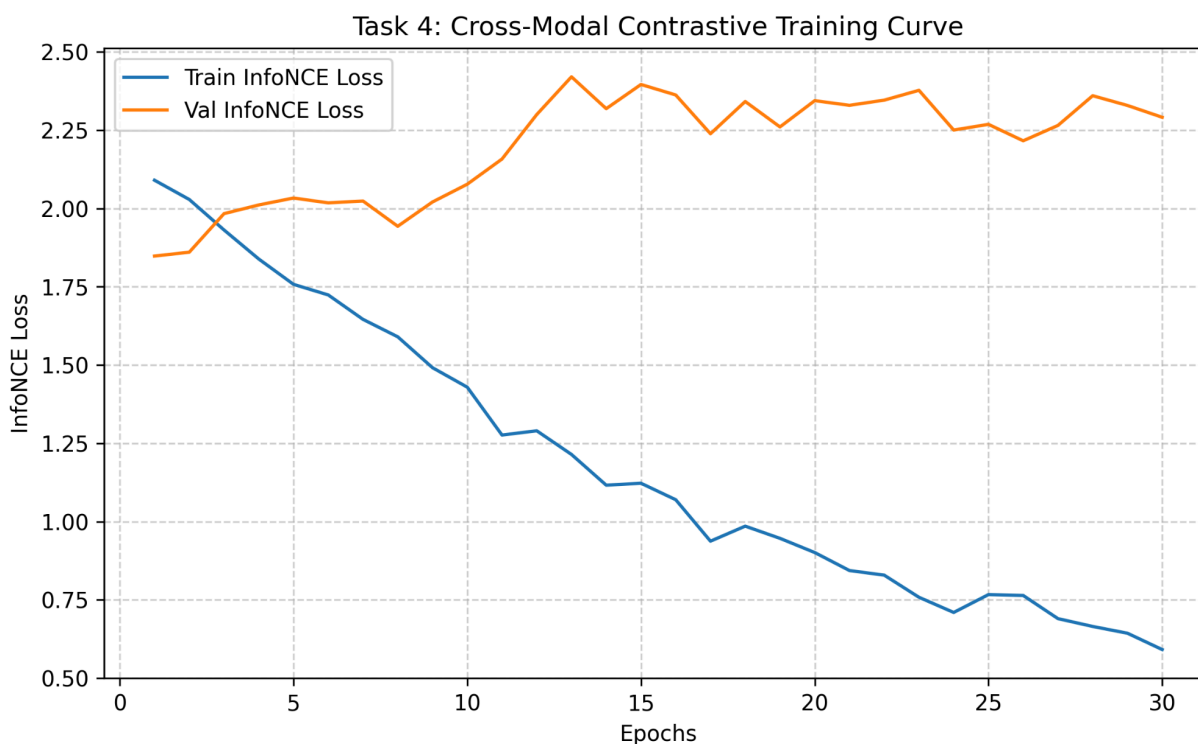


FIGURE: Contrastive training loss curve

6.3 Discussion

R@5 improved from 10.71% to 21.43% (a 2x improvement) and qualitative top-3 match rate improved from 1-of-10 to 4-of-10 after the hyperparameter correction, confirming the fix meaningfully improved the learned embedding space. With only 271 training pairs, this is a genuinely difficult contrastive learning setting: InfoNCE relies on a sufficient number of in-batch negatives to shape a discriminative embedding space, and small batch sizes on a small dataset limit this signal. Retrieved audio clips still showed some repetition across different query captions, suggesting the embedding space, while improved, has not fully separated all 271 tracks. Given Task 4's optional/bonus status and the CPU-only training constraint,

these results are reported as a proof-of-concept demonstrating the contrastive learning pipeline functions correctly and responds appropriately to hyperparameter tuning, rather than as a fully converged retrieval system.

Ten qualitative caption-to-audio retrieval examples, including top-3 retrieved clips and similarity scores, are provided in results/task4_qualitative_examples.txt.

7. Overall Discussion and Limitations

- Dataset scale: all experiments used CPU-only training on a laptop-scale dataset (271-635 tracks depending on task), well below the scale typically used for fine-tuning BERT or training GNNs from scratch; this is the primary driver of the overfitting observed in Tasks 1, 3, and 4.
- Lyrics quality: lyrics were ASR-derived (Whisper) rather than ground truth, introducing transcription noise, particularly for instrumental-heavy genres and tracks.
- Chord recognition simplicity: the template-matching approach models only major/minor triads and cannot represent extended, altered, or non-diatonic harmony, motivating the exclusion of classical and jazz from the dataset.
- Consistent cross-task finding: timbral/production information (as captured by mel-spectrograms) is a substantially stronger genre signal than harmonic movement alone (as captured by chord-transition graphs) in this dataset, though fusing chord-graph structure with lyrical semantics does improve on either single modality (Task 3).
- Reggae was a consistent weak point across every Task 3 model variant, warranting further investigation with a larger or more balanced dataset.

8. Conclusion

This project implemented and evaluated four progressively more complex approaches to modeling musical context: a text-only BERT baseline, a graph-only GNN with a CNN comparison baseline, a GNN-BERT fusion model with a four-way ablation study, and a cross-modal contrastive retrieval system. The central finding is that cross-attention fusion of graph and text representations outperforms either modality alone, validating the project's core hypothesis, while also revealing that spectrogram-based timbral features remain a stronger standalone genre signal than chord transition structure. All code, preprocessed graph samples, evaluation tables, and plots are included in the accompanying repository.

References

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